A six-equation two-pressure code against the ECCSEL CO_2 depressurization tests: the nucleation onset, the transfer rate, and what an HRM-type source shows

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23 September 2026 — draft for discussion, revised

Purpose

Log, Hammer, Deng, Austegard, Hafner and Munkejord (*Int. J. Multiphase Flow* 170 (2024) 104624) published seven full-bore depressurizations of dense liquid-like CO_2 in the ECCSEL pipe, with pressure traces at 100 kHz near the open end, and compared them with a homogeneous equilibrium model (HEM) and a pT-relaxed homogeneous relaxation model (HRM*) whose single relaxation time θ was fitted per test. This note runs three of those tests with CAMR, a six-equation Pelanti–Shyue two-pressure code, and reports three things: what the code's closure gets right and wrong against the measurements; that replacing the code's fixed nucleation margin by the classical-nucleation-theory onset reproduces the warm test's first level with nothing fitted; and what happens when the paper's source law is placed inside the two-pressure model. It is a basis for discussion, not a validation claim: the runs are one-dimensional, adiabatic and frictionless over the last 3–4 m of the pipe and the first 10 ms, and the transfer rate behind the onset remains the open closure.

1 The comparison

Tests. Test 19 (124.7 bar, 10.2 °C; the paper's $\theta = 2.5$ ms), Test 23 (121.9 bar, 31.5 °C; $\theta = 0.55$ ms; held out until the onset law was fixed) and Test 24 (115.6 bar, 35.8 °C; $\theta = 0.14$ ms). Sensors PT201–PT206 at 0.08, 0.18, 0.28, 0.484, 0.8 and 1.599 m from the open end; uncertainty about 0.6 bar.

CAMR. Six-equation model with instantaneous mechanical relaxation, a coupled pressure–temperature–Gibbs relaxation operator (X3) whose mass transfer uses a statistical-rate-theory (SRT) rate with an interfacial-area prefactor $\Sigma \propto (\alpha_2 + \delta)\alpha_1/D$ ($\delta = 0.01$, morphology length D, default 0.1 m), second-order wave-propagation hydrodynamics, 6.25 mm cells (the paper: 6.17 mm). A single-phase liquid below saturation acquires vapour through a *nucleator*: a smooth window in pressure that opens at $(1-m)P_{\mathrm{sat}}(T)$ and seeds vapour to a volume fraction of 0.04; the committed fraction is solved implicitly in the window. Until this work the margin was a constant, $m = 0.1$. It is now the homogeneous-nucleation onset at the liquid's temperature,

$$J = N_\ell \sqrt{2\sigma/\pi m_1}\, \exp[-16\pi\sigma^3/(3k_B T\, \Delta P^2)] = 10^{12}\ \mathrm{m^{-3}s^{-1}}, \qquad \Delta P = (P_{\mathrm{sat}} - P)(1 - \rho_v/\rho_\ell),$$

*Prepared with an AI assistant from the runs recorded on branch `co2-eos` of CAMR (tag `onset-2026-09-23`); every number is from a recorded run.

with the paper's Appendix A threshold and $\sigma(T) = 0.07863\,(1 - T/T_c)^{1.254}$ N/m (Mulero et al. 2012), capped at the former constant: $m(T) = \min(m_{\mathrm{CNT}}, 0.1)$. Nothing is fitted; the cap is the heterogeneous (wall) floor the cold data require. The two runs of each test below are "fixed window" (the former default) and "CNT onset" (the present one). The outlet is a characteristic face with an outflow fan; the far end of the truncated domain is a wall the wave never returns from. EOS: GERG-2008 for every run shown; Peng–Robinson for the parameter scans of Fig. 5.

The paper's model curves. The HEM and HRM* pressure at the 8 cm sensor for Tests 24, 19 and 23 were recovered from the vector paths of the paper's Figs. 13(b) and 15 (`digitise_paper.py`: `pdftocairo` to SVG, each panel calibrated by least squares on its own measured curve against the public dataset, residuals 0.02–0.07 bar). They carry the authors' plotting decimation (24–36 points over 15–30 ms) and are otherwise their curves; they are shown in Fig. 4 and are the basis of the 8 cm comparison. The authors' model includes wall friction and heat transfer and the whole pipe; CAMR has neither. At 8 cm and within 10 ms that difference is small.

The HRM-source probe. A switch replaced the nucleator by the paper's source, $\Gamma = \rho\,(x_{g,\mathrm{sat}} - x_g)/\theta$, active in every cell whose liquid is below $P_{\mathrm{sat}}(T)$, with $x_{g,\mathrm{sat}}$ the equilibrium quality at the cell's (ρ, e) on the EOS's own dome, the SRT transfer switched off, and the instantaneous mechanical and thermal relaxation kept — the pT-relaxed assumption of HRM* inside the two-pressure model. A probe of one question (Fig. 3, PR), not a proposed closure; it stays in the code as a probe key (`ps_flash_hrm`).

2 Results

Table 1: Numbers behind the figures (GERG unless marked PR). "Level" is the first plateau after the fast drop; for Test 23 the measured value is the bottom of the dip, not a plateau. HRM* rows are the paper's statements about its own model; its traces are not reproduced here.

		wave speed 0.8→3.2 m [m/s]	level, 0.8 m [bar]	dip → max, 0.8 m [bar]	0.08 m at +5 ms [bar]
Test 24	measured	370	63.8	flat	40.4
	HRM*, $\theta = 0.14$ ms	—	—	no plateau; drop too slow	overestimates
	CAMR, fixed window	366	60.0	flat	36.9 at 10 ms (meas. 37.2)
	CAMR, CNT onset	366	64.0	flat	36.5
	CAMR PR, fixed window, $D = 2$ mm	—	57.1	flat	41.2
	CAMR PR, HRM source, 0.14 ms	—	62.0	flat	30.0
Test 23	measured	441	42.9 (dip)	53 → 43 → 49	45.2
	CAMR, fixed window	419	55.4	55.3 → 47, no recovery	34.9
	CAMR, CNT onset	419	56.7	56.6 → 47, no recovery	34.9
Test 19	measured	585	34.2	32.5 → 34.6	21.8
	HRM*, $\theta = 2.5$ ms	—	—	captured (fitted); recovery "more abrupt" than measured	underestimates
	CAMR, fixed window	592	35.6	flat	25
	CAMR, CNT onset	592	35.6 (identical)	flat	25
	CAMR PR, HRM source, 2.5 ms	—	—	12.7 → 30.3	18.4
	CAMR PR, HRM source, 0.5 ms	—	—	31.8 → 34.8	24.8

What the two-pressure model gets right without tuning. The head of the depressurization wave travels at the measured speed (within 1.2 % at both ends; 5 % low on Test 23), and with the CNT onset the first level after the fast drop is within 0.2 bar of the measurement warm (64.0 vs.

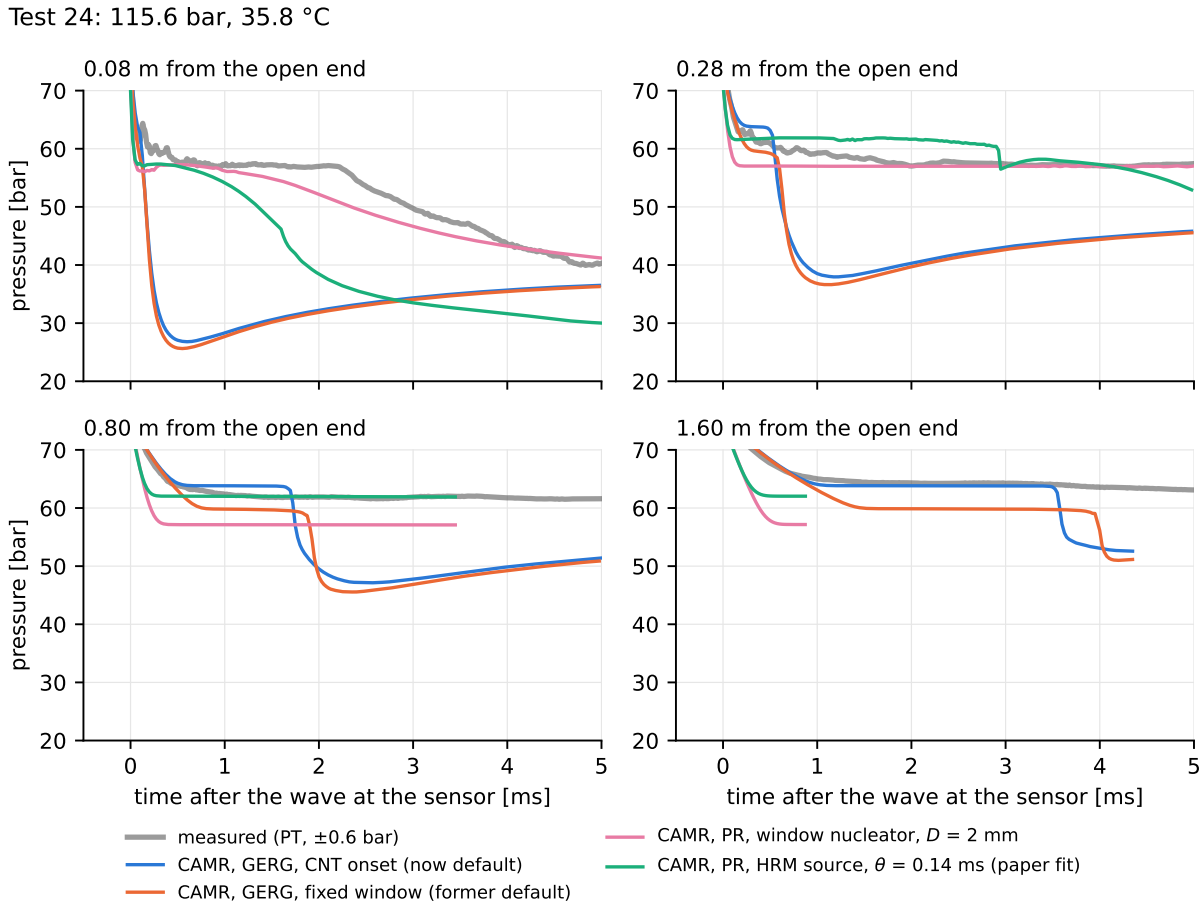


Figure 1: Test 24 (warm). Every trace is shifted so that its 40 bar drop at the sensor is $t = 0$ (the rupture disk opens over milliseconds; the code opens instantly), so absolute times are not comparable while levels and times after the wave are. The CNT onset lifts the first level from 0.92 to 0.98 P_{sat}; what follows it, the fall to 48 bar at 2 ms, is the transfer rate and is unchanged. The pink trace is the fixed window with the SRT morphology length reduced from 0.1 m to 2 mm.

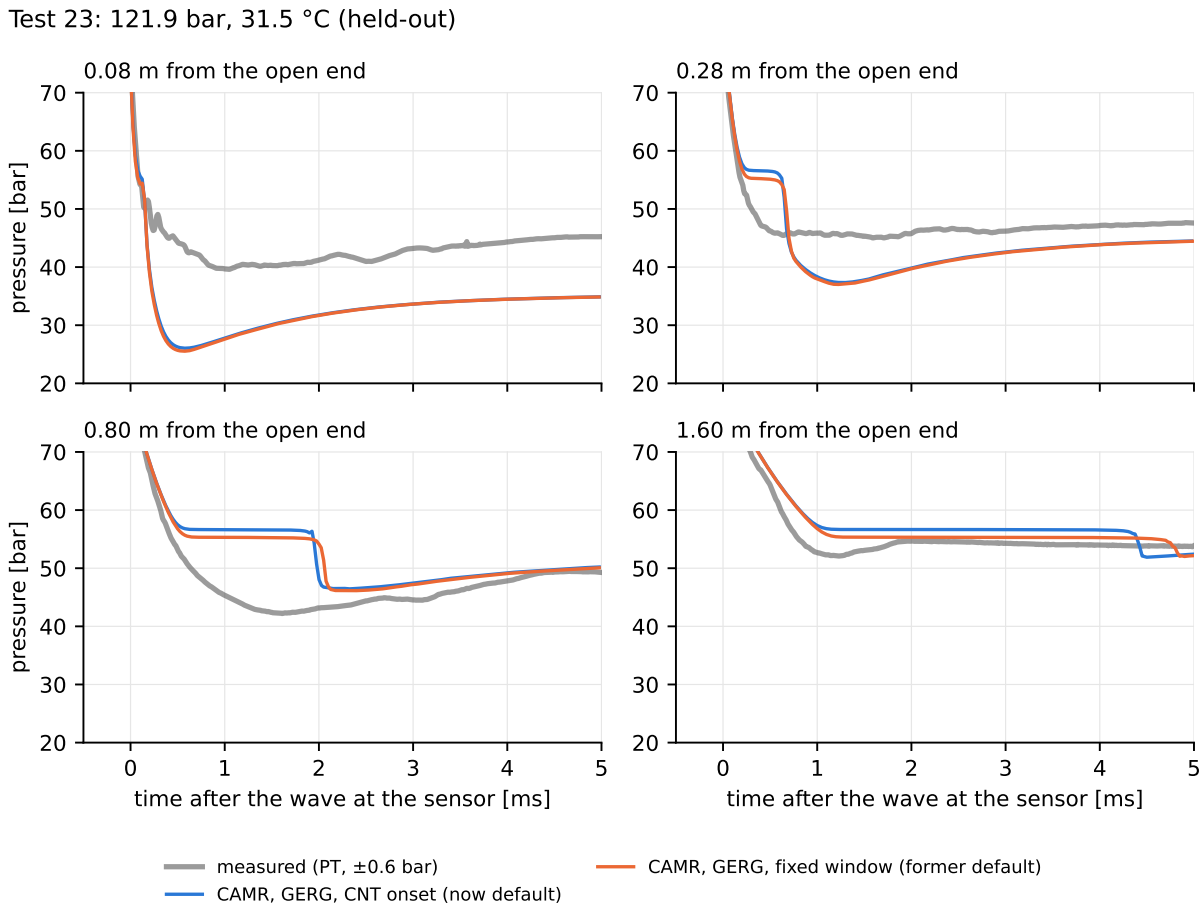


Figure 2: Test 23 (31.5 °C, held out). The measurement has no plateau near saturation: it falls 13 bar past the onset and recovers 6 bar. Both onset laws put the first level where they say (56.7 and 55.4 bar at 0.8 m); neither follows the dip, which the transfer rate governs. The CNT run leaves its plateau a millisecond earlier.

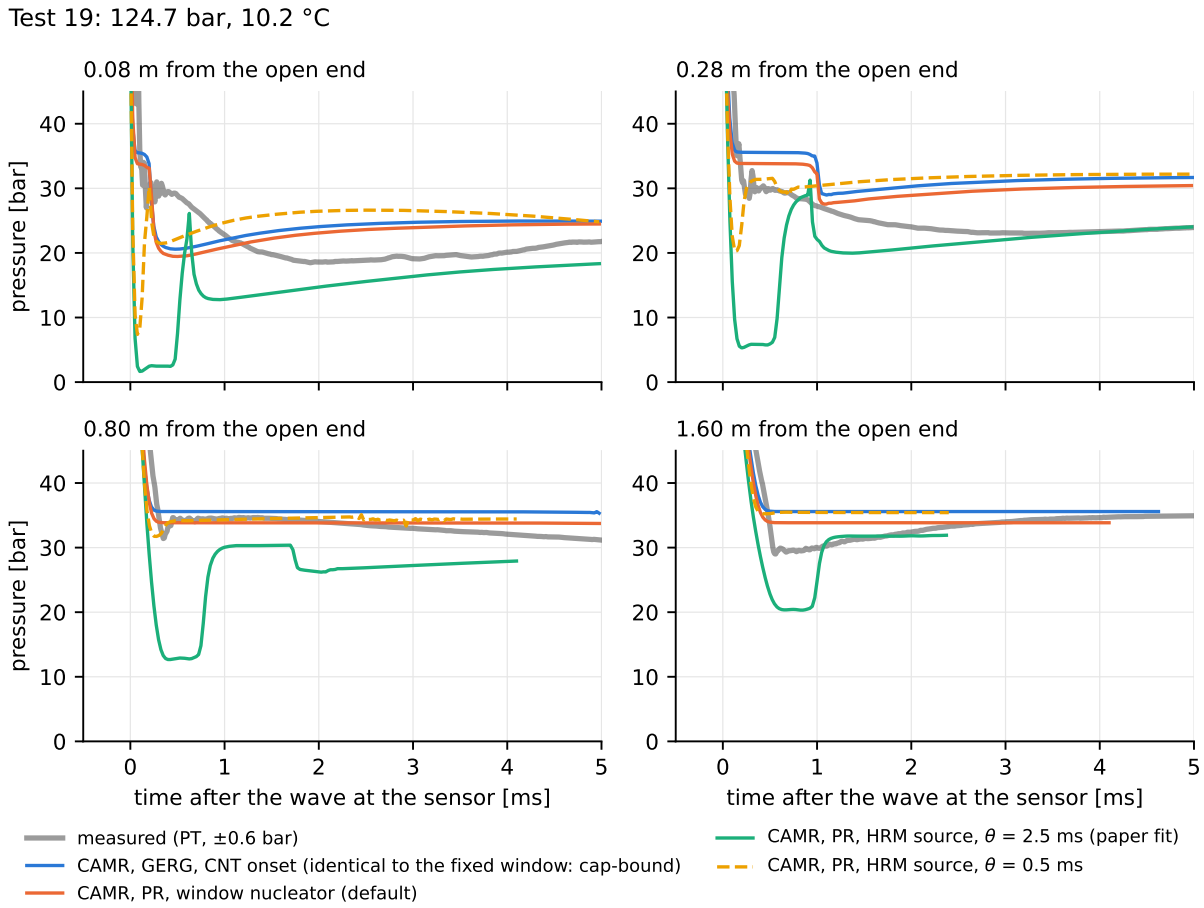


Figure 3: Test 19 (cold). The CNT-onset run is identical to the fixed window (the cap binds: the homogeneous limit would be a 63 % undershoot at 276 K against a measured 9 %). The green traces are the HRM-source probe (PR) at the paper's θ and at 0.5 ms: the undershoot and recovery appear at the measured times after the wave.

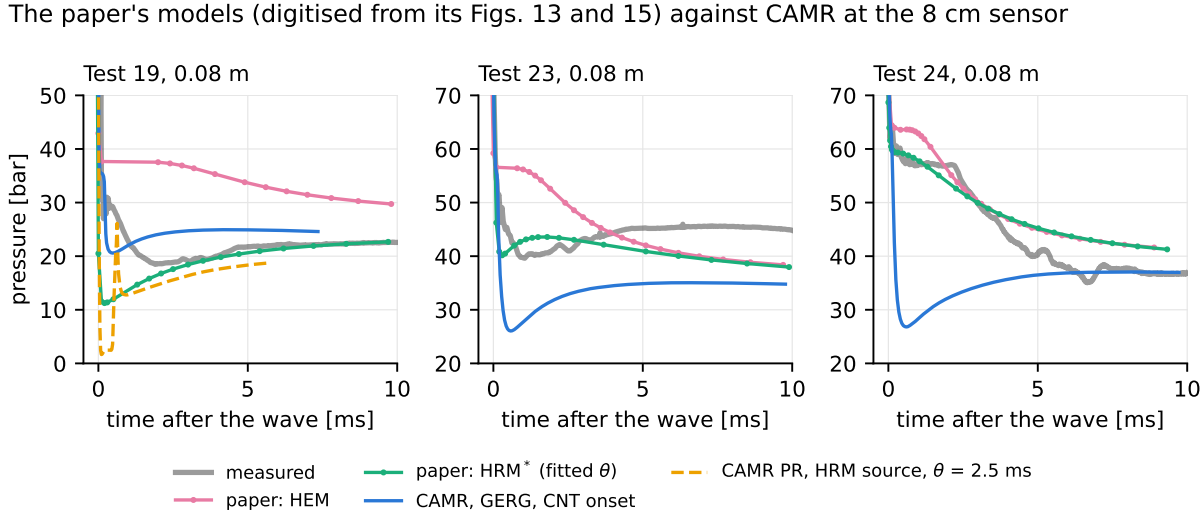


Figure 4: The 8 cm sensor, the paper's own comparison point: measured, the paper's HEM and HRM* (digitised), and CAMR with the CNT onset (GERG). Test 19 also shows CAMR's HRM-source probe at the paper's θ (PR). In the cold test CAMR is the closest of the three models to the measurement; in the two warmer tests it is 10–20 bar low after the first millisecond, where HRM* (fitted) sits on the data.

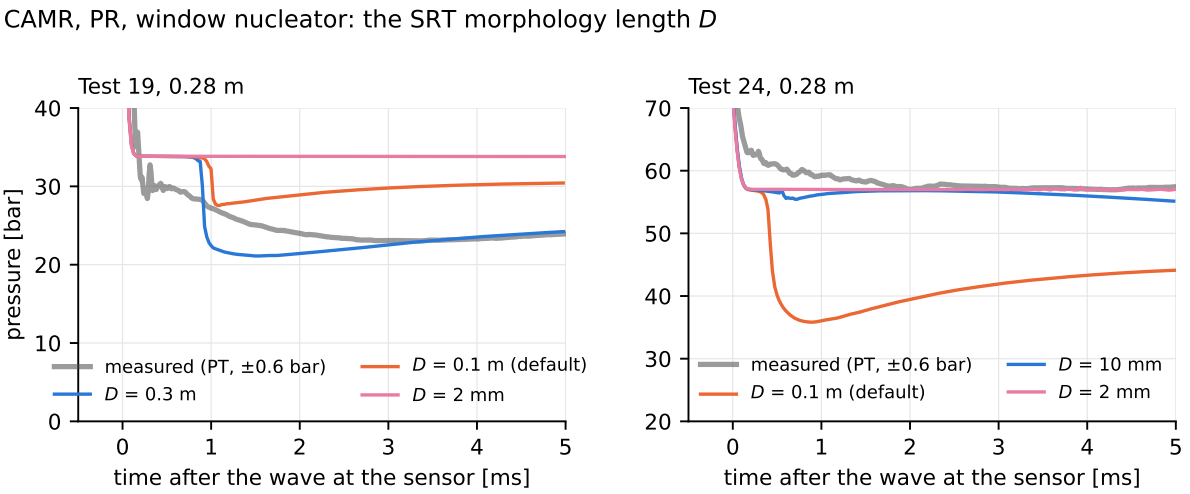


Figure 5: The morphology length D of the SRT rate, fixed window, PR, at the 0.28 m sensor: the cold test wants $D \approx 0.2$ m, the warm one $D \lesssim 2$ mm.

63.8 bar) and 1.4 bar cold (35.6 vs. 34.2). The warm number was predicted before the run (64.0 ± 0.5) from the law alone. The choked outlet pressure of the warm test at 10 ms is reproduced (36.5 vs. 37.2 bar).

What the onset law does and does not do. Near the critical point $\sigma \to 0$ and the homogeneous onset is at saturation ($m = 0.023$ at 298.5 K), which is what Test 24 shows. Far from it the homogeneous limit is unreachable in a pipe (63 % at 276 K) and the cap governs; the cold onset is therefore *not predicted* by this law, only left at the former constant, which the data happen to put near 9–10 %. A wall-nucleation law would replace the cap; that needs a site density and a contact angle the data do not supply.

What the transfer rate does not do. Behind the first level the code's SRT rate at the default morphology length is far too slow warm (Fig. 1: 48 bar at 2 ms against 62) and somewhat too fast cold; the two tests bracket D at 0.2 m and $\lesssim 2$ mm (Fig. 5), a range of 100 against the paper's 18:1 in θ. Test 23 is the sharpest statement of the same deficiency: at 31.5 °C the onset is right and the measured pressure still falls 13 bar below it and recovers 6, while the code holds its plateau (Fig. 2). There is no undershoot-and-recovery in any CAMR run, because the window pins the liquid at the onset and nothing keeps producing vapour once the pressure rises; and after the first drop every CAMR configuration we tried is stationary or monotone at the near-exit sensors, where the measurement falls a second time over 1.5–3 ms and recovers.

Against the paper's models at 8 cm (Fig. 4). Test 19: HEM sits at 37 bar from the first millisecond (measured 19–22); HRM* dips to 11 bar and recovers to 23 by 10 ms; CAMR dips to 21 and settles at 25 — the closest of the three to the measured 19–22, with no fitted constant, though without the measured shape. Test 23: HEM 56 bar, HRM* 40 recovering to 44 (measured 40–47), CAMR 26 recovering to 35 — 10 bar low. Test 24: HEM 64, HRM* 59 (measured 57–58 for 3 ms, then falling to 40), CAMR 27 recovering to 37 — CAMR reaches the measured 10 ms value but is 20–30 bar low in between. The pattern is the transfer-rate deficiency of Fig. 1 seen at the outlet: in the warm tests the code's vapour production behind the onset is too slow and the outlet cell decompresses far below the data before the SRT transfer catches up; in the cold test the same slowness happens to sit where the measurement is. HRM* is on the data in the warm tests because θ was fitted there.

What the HRM source inside the two-pressure model does. With the paper's source law the code produces the undershoot and the recovery (Fig. 3), at the measured time after the wave (0.40 and 0.82 ms against 0.33 and 0.79 ms at 0.8 and 1.6 m). So the two-pressure structure and the trace-phase treatment are not the obstacle; the feature needs a source that starts at the saturation line, is at full strength with no vapour present, and keeps producing while the pressure recovers. A transported interfacial-area density was also tried, coupled into the SRT rate, and refuted: it only rescaled the transfer and gave no recovery.

Two things we would like to discuss.

1. With the paper's source law at the paper's $\theta = 2.5$ ms, our probe at 8 cm falls to the vent's floor (2 bar at 1 bar ambient) and recovers to 18 by 5 ms, with a compression spike on the way up; the paper's HRM* at the same sensor stops at 11 bar and recovers to 23 (Fig. 4). A scan of θ (0.25–5 ms) and of the ambient pressure (1–6 bar) showed that the ambient sets only the floor, and the level after 2 ms is set by θ alone, about 2.5 bar lower per millisecond of θ at every sensor; our probe reaches HRM*'s dip and late level at $\theta \approx 1$ ms. So at equal θ our source is two to three times slower than the authors', which is what $x_{g,\mathrm{sat}}$ at fixed (ρ, e) against $x_{g,\mathrm{eq}}$ at (p, h) would do; we would like to confirm which quality HRM* relaxes toward. Both codes overshoot the measured dip, which is the paper's own observation that HRM* "cannot fit the pressure undershoot without providing a too strong superheat".

2. The onset near the critical point is homogeneous CNT with no free constant, and the cold onset is a wall property; between them Test 23 shows the transfer, not the onset, setting the dip. We built and tested a conduction-limited growth law (Plesset–Zwick, a bubble number density n as its one constant) as the rate below and above the presence threshold: at $n \approx 10^{13}\,\mathrm{m}^{-3}$ it puts Test 19's level after onset on the data at 8 and 28 cm, but the pressure is then stationary — no second decline, no recovery. A rupture-disc opening law, the vent's choking criterion (which did expose an over-expanded exit cell in our vent, Wood Mach 1.46, cured by ending the fan at the Bernoulli choking point) and thermal non-equilibrium between the phases were tested and refuted for the same feature. What none of our configurations shows is the slow decline the authors' HEM has at 8 cm (37 to 27 bar over 10 ms, the equilibrium two-phase fan), and the measured second decline looks like that fan, faster and deeper, followed by the non-equilibrium recovery. Our question is therefore: in HRM*, what produces the slow decline of Test 24 at 8 cm (their Fig. 13b) at $\theta = 0.14$ ms, and does HRM* produce one for Test 19 at any θ?

Caveats. One-dimensional, no wall friction or heat transfer, no disk-opening model, adiabatic; the last 3–4 m of the pipe only, so nothing here speaks to the plateau after 10 ms. The measured traces are the authors' public dataset (doi 10.5281/zenodo.7669536, CC BY 4.0).